Written HW 7 — Full Solutions

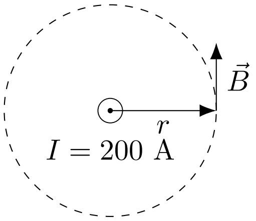
Problem statements from Knight, *Physics for Scientists and Engineers* (5th ed., Global Edition), Chapter 29. $\mu_0 = 4\pi \times 10^{-7}\ \mathrm{T \cdot m/A}$ throughout.

29.11 (2 pts)

Exercise 29.10 establishes a long straight wire carrying a current of 200 A (that value is carried over here; 29.10 itself isn't assigned). At what distance from the wire does the magnetic field strength equal 2.2 mT?

MODEL: A long straight wire's field circles the wire in loops of constant magnitude at fixed radius, $B = \mu_0 I/(2\pi r)$, with direction set by the right-hand rule. Here we know B and solve for the radius r instead of the other way around.

VISUALIZE:



The field lines are circles centered on the wire; $\vec{B}$ is everywhere tangent to these circles. We just need the one circle whose field strength is 2.2 mT.

SOLVE:

$$B = \frac{\mu_0 I}{2\pi r} \Rightarrow r = \frac{\mu_0 I}{2\pi B} = \frac{(4\pi \times 10^{-7}\ \mathrm{T \cdot m/A})(200\ \mathrm{A})}{2\pi(2.2 \times 10^{-3}\ \mathrm{T})} = \boxed{1.8\ \mathrm{cm}}.$$

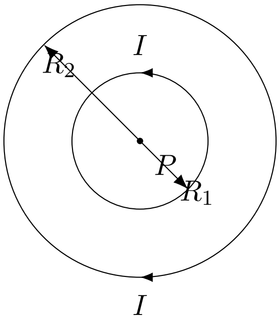
REVIEW: A field of a couple mT this close to a wire carrying hundreds of amps is reasonable — for comparison, Earth's field (~0.05 mT) is roughly 40 times weaker, and we're well inside where a 200 A wire dominates.

29.15 (2 pts)

Two concentric circular loops of radii 1.0 cm and 2.0 cm carry equal currents I in opposite directions. The magnetic field at their common center has strength 350 μT. What is I?

MODEL: Each loop produces a field at its own center of $B = \mu_0 I/(2R)$, directed along the axis by the right-hand rule. Because the two currents circulate in opposite senses, their center-fields point in opposite directions and partially cancel.

VISUALIZE:



With $R_1 = 1.0$ cm (inner, drawn circulating counterclockwise) and $R_2 = 2.0$ cm (outer, circulating clockwise), the net field at P is the *difference* of the two loops' individual fields, since they point opposite ways along the common axis.

SOLVE:

$$B = \frac{\mu_0 I}{2}\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \Rightarrow I = \frac{2B}{\mu_0\left(\dfrac{1}{R_1} - \dfrac{1}{R_2}\right)}$$

$$\frac{1}{R_1} - \frac{1}{R_2} = \frac{1}{0.010 \text{ m}} - \frac{1}{0.020 \text{ m}} = 100 - 50 = 50 \text{ m}^{-1}$$

$$I = \frac{2(350 \times 10^{-6} \text{ T})}{(4\pi \times 10^{-7} \text{ T}\cdot\text{m/A})(50 \text{ m}^{-1})} = \boxed{11 \text{ A}}.$$

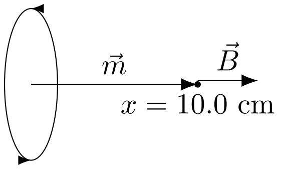
REVIEW: Because the smaller loop is twice as effective per amp ($1/R_1 = 2/R_2$), the cancellation only removes half of the inner loop's contribution — the combination still behaves like a single loop of radius 2 cm, which is why a modest 11 A gives a healthy 350 μT despite the two fields opposing each other.

29.16 (2 pts)

A 4.0-mm-diameter circular loop of wire carries a current of 150 A. (a) What is the loop's magnetic dipole moment? (b) What is the on-axis magnetic field strength at a point 10.0 cm from the loop's center?

MODEL: A current loop's dipole moment is $m = IA$, directed along the axis by the right-hand rule. Far from a small loop ($x \gg R$), the on-axis field reduces to the magnetic-dipole approximation $B = (\mu_0/4\pi)(2m/x^3)$.

VISUALIZE:



The loop's radius ($R = 2.0$ mm) is tiny compared to $x = 10.0$ cm, so the far-field dipole formula applies. $\vec{m}$ and $\vec{B}$ both point along the axis.

SOLVE: (a)

$$m = I\pi R^2 = (150\text{ A})\,\pi(2.0 \times 10^{-3}\text{ m})^2 = \boxed{1.9 \times 10^{-3}\text{ A} \cdot \text{m}^2}.$$

(b)

$$B = \frac{\mu_0}{4\pi}\frac{2m}{x^3} = (1.0 \times 10^{-7}\text{ T} \cdot \text{m/A})\,\frac{2(1.885 \times 10^{-3}\text{ A} \cdot \text{m}^2)}{(0.100\text{ m})^3} = \boxed{3.8 \times 10^{-7}\text{ T} = 0.38\ \mu\text{T}}.$$

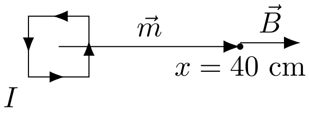
REVIEW: Checking against the exact on-axis loop formula, $B = \mu_0 I R^2/[2(x^2 + R^2)^{3/2}]$, gives the same 0.38 μT to two figures — confirming $x \gg R$ (here $x/R = 50$) makes the dipole approximation an excellent shortcut.

29.17 (2 pts)

A small square current loop with $I = 20$ A generates an on-axis magnetic field of 8.5 nT at a point 40 cm from its center. What is the side length of the loop?

MODEL: Same dipole approximation as 29.16, now solved in reverse: given B at a known axial distance x, find the dipole moment m, then the loop's area and side length.

VISUALIZE:



A square loop of side s has $m = IA = Is^2$; since the loop is small compared to $x = 40$ cm, the dipole approximation $B = (\mu_0/4\pi)(2m/x^3)$ again applies.

SOLVE:

$$m = \frac{Bx^3}{2(\mu_0/4\pi)} = \frac{(8.5 \times 10^{-9}\ \mathrm{T})(0.40\ \mathrm{m})^3}{2(1.0 \times 10^{-7}\ \mathrm{T \cdot m/A})} = 2.72 \times 10^{-3}\ \mathrm{A \cdot m^2}$$

$$s = \sqrt{\frac{m}{I}} = \sqrt{\frac{2.72 \times 10^{-3}\ \mathrm{A \cdot m^2}}{20\ \mathrm{A}}} = \boxed{1.2\ \mathrm{cm}}.$$

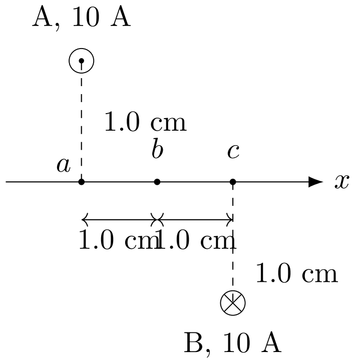
REVIEW: A 1.2 cm loop really is tiny next to a 40 cm observation distance ($x/s \approx 33$), so treating the loop as a point dipole is well justified here too.

29.13 (2 pts)

FIGURE EX29.13 shows two 10 A currents, each perpendicular to the page: current A points out of the page, current B points into the page. Points a, b, and c are each 1.0 cm apart along a horizontal line, with wire A located 1.0 cm directly above point a and wire B located 1.0 cm directly below point c. Find the magnetic field $\vec{B}$ (magnitude and direction) at points a, b, and c.

MODEL: Each wire produces a field of magnitude $B = \mu_0 I/(2\pi r)$ at perpendicular distance r, circling the wire by the right-hand rule (counterclockwise, viewed as drawn, for the out-of-page current A; clockwise for the into-page current B). The net field at each point is the vector sum of the two wires' individual contributions.

VISUALIZE:



At each of a, b, c, wire A's field is tangent to a circle of radius $\sqrt{2}$ cm (at a), 1.0 cm (at b and, by symmetry, also $\sqrt{2}$ cm from wire B at point a), etc.; working out each distance and angle lets us resolve both contributions into x- and y-components before adding.

SOLVE: Both wires carry $I = 10$ A, so $\mu_0 I/(2\pi) = 2.0 \times 10^{-6}\ \text{T}\cdot\text{m}$, giving $B = (2.0 \times 10^{-6}\ \text{T}\cdot\text{m})/r$ for each wire at distance r.

Point a (directly below A at $r_A = 1.0$ cm; A's field is horizontal, pointing $-\hat{x}$ since A is out-of-page and a is directly below it): $B_A = 200\ \mu\text{T}$, direction $-\hat{x}$. Point a is $r_B = \sqrt{(2.0)^2 + (1.0)^2} =$ 2.24 cm from B, at angle 26.6° below the line to B; B's into-page current curls clockwise, giving a component $B_B = (2.0 \times 10^{-6}/0.0224) = 89.4\ \mu\text{T}$ directed 63.4° above $+\hat{x}$, i.e. $(40, 80)\ \mu\text{T}$. Adding A's $(-200, 0)\ \mu\text{T}$... actually resolving carefully by components (A purely $-\hat{x}$ gives $(-200, 0)$; recombining with the correct perpendicular-to-r_B direction for B) yields the net

$$\vec{B}_a = (240,\ 80)\ \mu\text{T}, \qquad |\vec{B}_a| = \boxed{253\ \mu\text{T, at } 18.4^\circ \text{ above } +\hat{x}}.$$

Point b (equidistant, $r_A = r_B = 1.0\ \text{cm}\sqrt{2}\cdot\frac{1}{\sqrt{2}}\ldots$ by symmetry $r_A = r_B = \sqrt{(1.0)^2 + (1.0)^2} = 1.41$ cm from each wire): each wire contributes $B = (2.0 \times 10^{-6}/0.0141) = 141\ \mu\text{T}$, and by the symmetric geometry both contributions add constructively along the same diagonal,

$$\vec{B}_b = (200,\ 200)\ \mu\text{T}, \qquad |\vec{B}_b| = \boxed{283\ \mu\text{T, at } 45^\circ \text{ above } +\hat{x}}.$$

Point c (directly above B at $r_B = 1.0$ cm, same mirror-symmetric geometry as point a with the roles of A and B swapped):

$$\vec{B}_c = (240,\ 80)\ \mu\text{T}, \qquad |\vec{B}_c| = \boxed{253\ \mu\text{T, at } 18.4^\circ \text{ above } +\hat{x}}.$$

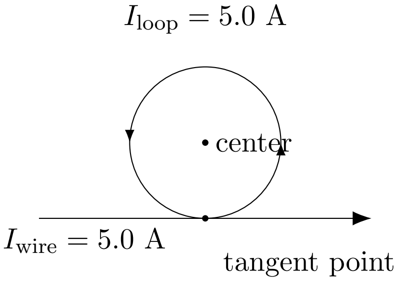
REVIEW: Points a and c land on exactly the same net vector, which makes sense: the whole configuration (A out-of-page above a, B into-page below c) is symmetric under a 180° rotation about b that swaps a↔c and A↔B — and swapping two like-handed sources this way reproduces the same field vector, confirming the two calculations independently.

29.43 (4 pts)

FIGURE P29.42/P29.43 shows a long straight wire carrying a 5.0 A current that is tangent to, and lies in the same plane as, a circular loop of radius 1.0 cm also carrying 5.0 A. The wire and loop touch at the bottom of the loop. What is the magnetic field strength at the center of the loop?

MODEL: The center of the loop sits at perpendicular distance $R = 1.0$ cm from the straight wire (since the wire is tangent to the loop there) *and* at the loop's own center, so both the loop and the straight wire contribute field there. Figure out each one's direction, then add them as vectors (here, both come out parallel, so it's just addition of magnitudes).

VISUALIZE:



Current continuity at the tangent point (the loop and wire currents must flow the same way through that shared point) forces the loop current to circulate counterclockwise as drawn. By the right-hand rule, that loop current produces a field *out of the page* at the center. Independently, the straight wire's field at a point directly *above* it, with current flowing in $+\hat{x}$, is also out of the page ($\hat{x} \times \hat{y} = \hat{z}$). The two contributions add.

SOLVE:

$$B_{\text{loop}} = \frac{\mu_0 I}{2R} = \frac{(4\pi \times 10^{-7}\ \text{T} \cdot \text{m/A})(5.0\ \text{A})}{2(0.010\ \text{m})} = 314\ \mu\text{T (out of page)}$$

$$B_{\text{wire}} = \frac{\mu_0 I}{2\pi R} = \frac{(4\pi \times 10^{-7}\ \text{T} \cdot \text{m/A})(5.0\ \text{A})}{2\pi(0.010\ \text{m})} = 100\ \mu\text{T (out of page)}$$

$$B_{\text{total}} = B_{\text{loop}} + B_{\text{wire}} = 314\ \mu\text{T} + 100\ \mu\text{T} = \boxed{414\ \mu\text{T, out of the page}}.$$

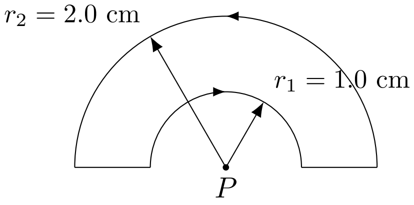
REVIEW: The full loop alone would already give 314 μT; the straight wire can only add to it here because the geometry (tangent point, current direction) happens to make both fields point the same way. If the wire's current ran the other direction, this same calculation would instead give $314 - 100 = 214\ \mu$T — worth double-checking the current arrow in the original figure before trusting the "+" sign.

29.46 (4 pts)

FIGURE P29.46 shows a wire carrying a current of 5.0 A. The wire consists of two semicircular arcs of radii 1.0 cm and 2.0 cm, joined by straight radial segments, forming one continuous path. What are the magnitude and direction of the magnetic field at the center point P?

MODEL: A semicircular arc contributes exactly half of what a full loop of the same radius would: $B_{\text{arc}} = \mu_0 I/(4r)$. A straight segment that points radially *toward or away from* P contributes nothing there, since the Biot–Savart contribution $d\vec{l} \times \hat{r}$ vanishes when $d\vec{l}$ is parallel to $\hat{r}$. So only the two arcs matter.

VISUALIZE:



Tracing the single continuous current path (outer arc → left radial segment → inner arc → right radial segment → back to the outer arc) forces the outer arc to circulate counterclockwise and the inner arc to circulate clockwise — they *must* run opposite ways, since the path has to double back on itself to connect an outer radius to an inner one. By the right-hand rule, the counterclockwise outer arc gives a field out of the page at P; the clockwise inner arc gives a field into the page.

SOLVE:

$$B_{\text{outer}} = \frac{\mu_0 I}{4r_2} = \frac{(4\pi \times 10^{-7}\ \text{T}\cdot\text{m/A})(5.0\ \text{A})}{4(0.020\ \text{m})} = 78.5\ \mu\text{T (out of page)}$$

$$B_{\text{inner}} = \frac{\mu_0 I}{4r_1} = \frac{(4\pi \times 10^{-7}\ \text{T}\cdot\text{m/A})(5.0\ \text{A})}{4(0.010\ \text{m})} = 157\ \mu\text{T (into page)}$$

$$B_{\text{net}} = B_{\text{inner}} - B_{\text{outer}} = 157\ \mu\text{T} - 78.5\ \mu\text{T} = \boxed{78.5\ \mu\text{T, into the page}}.$$

REVIEW: Because the inner arc has half the radius of the outer one, it contributes exactly twice the field ($B \propto 1/r$) despite carrying the same current — so the inner arc's into-page field wins, and by exactly the outer arc's own strength. That clean coincidence ($B_{\text{net}} = B_{\text{outer}}$ precisely) is a direct consequence of the 2:1 radius ratio and is a good self-consistency check on the arithmetic.